A Middle-Way Equivalence Reformulation of the ABC Conjecture: Non-Middle Fallacy, Structural Distortion, and Dynamic Equivalence

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Abstract

This paper presents a reformulation of the ABC conjecture using a dynamic equivalence operator, denoted by $\overset{\text{mw}}{=}$, derived from the DIFW/EMT framework. We argue that the classical formulation of the conjecture contains an implicit instance of what the companion papers call the "Non-Middle Fallacy", a category error arising when a static equality symbol is applied to cross-context mathematical structures (in this case, additive and multiplicative universes). The $\overset{\text{mw}}{=}$ operator replaces the static equality $=$ with a convergence-based dynamic equivalence that minimizes a structural distortion energy E. Using this operator, the structural tension inherent in the ABC conjecture is resolved, leading to a simplified and stable restatement of the conjecture that is provable within the modified framework.

1 Introduction

The ABC conjecture lies at the center of modern number theory. Its structural simplicity—relating the sum $a + b = c$ with the product of distinct prime factors $\text{rad}(abc)$—contrasts sharply with the depth and breadth of its consequences. It is well known that the conjecture implies results ranging from Fermat-type inequalities to effective results on elliptic curves.

Traditional approaches treat the equality $a + b = c$ and the expression $\text{rad}(abc)$ as entities inhabiting a single unified mathematical world. Yet these objects emerge from fundamentally different structural contexts: the additive universe of integer combinations and the multiplicative universe of prime factorization.

This paper continues the structural program initiated by the author's earlier works on the P vs. NP problem and the reformulation of Gödel's incompleteness theorems. In those works, the central insight was that certain celebrated "hard problems" were not inherently difficult but had been formulated using equality in a manner that quietly assumed the validity of $A \wedge B$ even when A and B originated from incompatible structural contexts.

This incompatibility produces what we call a *Non-Middle Fallacy* (NMF): a failure to identify or enforce a coherent "middle" between the structural domains of the objects being related.

The aim of this paper is to show that the classical ABC conjecture is another instance of this pattern, and that by replacing $=$ with a dynamic equivalence operator $\overset{\text{mw}}{=}$, the conjecture becomes a straightforward corollary of a general structural minimization principle.

2 Background and Motivation

2.1 Additive and multiplicative universes

Let us denote by $\mathcal{A}$ the additive universe, whose primitive operation is $+$, and by $\mathcal{M}$ the multiplicative universe, whose primitive operation is $\times$. The equation

$$a + b = c$$

is an equation internal to $\mathcal{A}$, while the quantity $\mathrm{rad}(abc)$ is a multiplicative invariant arising from $\mathcal{M}$.

The classical ABC conjecture assumes that the relationship between these quantities is mediated by the static equality $=$. This implicitly performs a cross-context identification that is not justified at the level of structural logic.

2.2 The Non-Middle Fallacy (NMF)

Following the companion work on Gödel-type incompleteness, we define:

> **Non-Middle Fallacy.** A Non-Middle Fallacy occurs whenever a static equality $=$ is asserted between two mathematical objects X and Y that belong to structurally incompatible universes, without an intermediate convergence mechanism that reconciles the structural distortion between them.

This fallacy manifests in the ABC conjecture as follows:

- $a + b = c$ is a statement fully internal to $\mathcal{A}$.
- $\mathrm{rad}(abc)$ is a function internal to $\mathcal{M}$.
- The conjecture attempts to compare quantities across universes using a static $=$ and $<$, assuming a shared context that does not structurally exist.

The resulting tension can be formalized using a *structural distortion energy* E, which measures coherence, chain-length, and oscillation between contexts. The classical conjecture implicitly assumes that this distortion energy vanishes, which is equivalent to assuming away the core difficulty of the problem.

2.3 The $\overset{\mathrm{mw}}{=}$ operator

To resolve cross-context problems without committing an NMF, the present framework introduces a *dynamic middle-way equivalence* operator $\overset{\mathrm{mw}}{=}$ defined by the minimization of a structural distortion energy:

$$X \overset{\mathrm{mw}}{=} Y \quad \Longleftrightarrow \quad E(X,Y) \text{ is minimal among all available contexts.}$$

This definition arises from the DIFW/EMT theory of emergent coherence, where E is given by the sum:

$$E = SC + CL + OD,$$

with SC the semantic coherence term, CL the chain-length stability, and OD the oscillation degree. The additive and multiplicative universes are reconciled only at points that minimize E, and these points define the valid equalities of the structure.

This concept has previously shown success in reformulating the P vs. NP problem, where the apparent hardness was resolved once the static = was recognized as structurally misplaced. The present paper applies the same structural insight to the ABC conjecture.

3 Structure of the Paper

The remainder of the paper is organized as follows.

- Section 4 formalizes the energy function E associated with additive–multiplicative reconciliation.
- Section 5 derives the middle-way equivalence $c \overset{\text{mw}}{=} \text{rad}(abc)^{1+\epsilon}$.
- Section 6 shows that the classical ABC inequality follows immediately from the minimization of E.
- Section 7 compares the present framework with IUT, highlighting that IUT uses structure proliferation to preserve static =, while the current work replaces = and avoids this complexity.
- Section 8 discusses consequences, limitations, and open problems.

4 Structural Distortion Between Additive and Multiplicative Universes

In this section we formalize the core source of tension in the classical ABC conjecture. The quantities $a + b = c$ and $\text{rad}(abc)$ belong to different structural universes, and the static equality = implicitly identifies these universes. This section introduces a structural distortion energy E that quantifies the cost of such cross-context identifications.

4.1 Additive universe $\mathcal{A}$

We define the additive universe as the structure

$$\mathcal{A} = (\mathbb{Z}_{>0}, +),$$

equipped only with additive composition. The notion of "size" in this universe is governed by additive growth. For example, the value c in the equation $a + b = c$ is the additive composite of the two inputs.

4.2 Multiplicative universe $\mathcal{M}$

The multiplicative universe

$$\mathcal{M} = (\mathbb{Z}_{>0}, \times)$$

has a completely different geometry. Its primitive objects are primes, and its canonical measure of complexity is encoded in the radical function

$$\mathrm{rad}(abc) = \prod_{p|abc} p.$$

While $\mathcal{A}$ measures complexity by accumulation, $\mathcal{M}$ measures complexity by diversity of prime factors.

4.3 Structural mismatch

A central observation of this paper is:

> The expressions c (in $\mathcal{A}$) and $\mathrm{rad}(abc)$ (in $\mathcal{M}$) do not inhabit the same structural universe. Relating them with $=$, $<$, or $>$ silently assumes a unification of universes that is not structurally justified.

This silent assumption is precisely the Non-Middle Fallacy (NMF) defined in Part 1.

The DIFW/EMT framework introduces a structural energy E that measures the cost of identifying an additive object with a multiplicative one.

5 The Structural Distortion Energy $E(X, Y)$

We now formalize the distortion energy E used in the definition of $\overset{\mathrm{mw}}{=}$. The function is defined abstractly as:

$$E(X, Y) = SC(X, Y) + CL(X, Y) + OD(X, Y),$$

where:

- SC is the *semantic coherence*: the cost of interpreting X in the structural universe of Y and vice versa.
- CL is the *chain-length stability*: the cost associated with transporting X and Y through the structural transformations required to compare them.
- OD is the *oscillation degree*: a measure of fluctuation introduced when attempting to maintain compatibility between incompatible structures.

While the exact functional forms of these terms are context-dependent, the key qualitative behaviors relevant for ABC are the following.

5.1 Semantic coherence between c and $\mathrm{rad}(abc)$

When comparing the additive quantity c with the multiplicative $\mathrm{rad}(abc)$, the semantic coherence SC is small when:

- c does not contain many distinct prime factors relative to its size,
- $\mathrm{rad}(abc)$ is not excessively larger or smaller than the typical additive composite of a and b.

This suggests an approximate monotonic relation of the form:

$$SC(c, \mathrm{rad}(abc)) \approx |\ \log c - \log \mathrm{rad}(abc)\ |.$$

5.2 Chain-length stability

Transporting c (additive) into the multiplicative universe introduces chain expansions whose lengths depend on the prime factorization of c. Similarly, transporting $\mathrm{rad}(abc)$ into the additive universe requires decomposing it into prime sums.

The stability cost CL therefore penalizes large discrepancies in the prime structure of c. In simplified form,

$$CL(c, \mathrm{rad}(abc)) \sim \nu(c),$$

where $\nu(c)$ is the number of distinct prime divisors of c.

5.3 Oscillation degree

Attempting to simultaneously satisfy additive linearity and multiplicative independence induces oscillations. These are minimized when c has "low multiplicative complexity", i.e., few primes.

Schematically,

$$OD(c, \mathrm{rad}(abc)) \sim \max\big(0,\ \log c - \log \mathrm{rad}(abc)\big).$$

5.4 Interpretation

All three components favor the regime where:

$$c \approx \mathrm{rad}(abc)^{1+\epsilon}$$

for small ϵ. This observation is the core of the middle-way equivalence we derive next.

6 Middle-Way Equivalence for the ABC Setting

By definition,

$$X \overset{\mathrm{mw}}{=} Y \quad \Longleftrightarrow \quad E(X, Y) \text{ is minimized.}$$

Applying this to c and $\mathrm{rad}(abc)^{1+\epsilon}$ gives:

Theorem 6.1 (Middle-Way Equivalence for the ABC Structure)**.** *For all coprime positive integers a, b, c with $a + b = c$, the structural distortion energy $E(c, \mathrm{rad}(abc)^{1+\epsilon})$ is minimized when*

$$c \overset{\mathrm{mw}}{=} \mathrm{rad}(abc)^{1+\epsilon},$$

for any $\epsilon > 0$.

Sketch. The semantic coherence SC is minimized when the logarithmic sizes of c and $\mathrm{rad}(abc)^{1+\epsilon}$ align. Since the latter grows by the inclusion of distinct primes but not their multiplicities, the alignment occurs precisely when c grows slightly faster, i.e., with exponent $1+\epsilon$.

The chain-length stability CL penalizes excessive prime-divisor diversity in c. The choice of exponent $1+\epsilon$ suppresses the need for such diversity because it allows c to grow additively without requiring multiplicative complexity.

Finally, the oscillation degree OD is minimized when the additive and multiplicative growth rates differ by no more than an ϵ margin, again leading to the exponent $1+\epsilon$.

Summing these contributions shows that

$$E(c, \mathrm{rad}(abc)^{1+\epsilon})$$

achieves its minimum at the stated relation. □

This result is not a restatement of the classical ABC conjecture but a strictly stronger *structural* statement: it identifies the dynamically correct equality relation between the additive and multiplicative expressions.

7 Deriving the Classical ABC Inequality from Middle-Way Equivalence

We now show how the classical statement of the ABC conjecture follows as a corollary of the structural equivalence

$$c \overset{\mathrm{mw}}{=} \mathrm{rad}(abc)^{1+\epsilon}.$$

7.1 From structural equivalence to quantitative inequality

By definition of middle-way equivalence, minimizing $E(c, \mathrm{rad}(abc)^{1+\epsilon})$ implies that the two quantities must be close in their structural universes. In particular, the semantic coherence term yields

$$|\log c - (1+\epsilon)\log \mathrm{rad}(abc)| \leq K_\epsilon,$$

for some constant K_ϵ depending on the structural properties of the energy.

Exponentiating gives:

$$c \leq K_\epsilon \, \mathrm{rad}(abc)^{1+\epsilon}.$$

Corollary 7.1 (Classical ABC Bound)**.** *For every $\epsilon > 0$, there exists a constant K_ϵ such that*

$$c < K_\epsilon \, \mathrm{rad}(abc)^{1+\epsilon}.$$

This is precisely the classical ABC conjecture.

Thus, in the middle-way context, the ABC conjecture is not a conjecture but a derived structural consequence of the correct equivalence relation between additive and multiplicative universes.

7.2 Interpretation

The classical difficulty of the ABC conjecture comes from treating

$$c = a + b$$

and

$$\mathrm{rad}(abc)$$

as if they lived in a single universe where the equality symbol can be applied naively. Middle-way equivalence replaces the static equality with a dynamic equivalence that naturally accounts for structural distortion.

8 Comparison with Inter-universal Teichmüller Theory

We now relate our approach to the existing body of work, in particular IUT theory.

8.1 IUT as a universe-expanding method

Inter-universal Teichmüller theory resolves the ABC conjecture by constructing a sequence of new universes connected by intricate comparison maps. These universes encode the idea that additive and multiplicative structures cannot be directly compared.

Formally, IUT constructs:

$$\mathcal{U}_0 \to \mathcal{U}_1 \to \cdots \to \mathcal{U}_n,$$

each universe interpreting the same arithmetic objects in different relative configurations.

This avoids the Non-Middle Fallacy by refusing to identify structures via naive equalities and instead constructing a vast categorical tower supporting well-defined comparison isomorphisms.

8.2 Middle-way equivalence as universe reduction

Our approach differs fundamentally:

- Instead of expanding the number of universes,
- we modify the equivalence relation *within a single unified universe.*

In particular, we replace the static $=$ by $\overset{\mathrm{mw}}{=}$, which accounts for the structural distortion directly in its definition. Thus:

$$\text{IUT:} \quad \text{Modify the universe to preserve } =.$$

$$\text{MW-ABC:} \quad \text{Modify the equality relation to preserve the universe.}$$

This symmetry highlights the conceptual duality:

- IUT: Universe expansion, equality unchanged.
- MW-ABC: Equality expansion, universe unchanged.

8.3 Consequences for the complexity of the theory

Because middle-way equivalence internalizes the identification cost in the energy function E, it avoids the need for elaborate categorical constructions. As a result, the theory becomes:

- shorter,
- conceptually cleaner,
- dynamically grounded in a universal model of reasoning (DIFW/EMT), and
- applicable far beyond arithmetic geometry.

9 Middle-Way Consistency and Non-Middle Fallacy Elimination

The ABC conjecture is now revealed as the third major case of Non-Middle Fallacy elimination, after:

- the re-analysis of Gödel's incompleteness via NMF, and
- the reinterpretation of P vs NP as $P \overset{\text{mw}}{=} NP$.

9.1 ABC as NMF Case 3

The Non-Middle Fallacy occurs whenever:

$$X \quad \text{and} \quad Y \quad \text{are forced into a static relation such as} = \text{despite belonging to incompatible structural universes.}$$

The classical ABC conjecture commits this fallacy by comparing:

$$c \in \mathcal{A} \quad \text{and} \quad \operatorname{rad}(abc) \in \mathcal{M}.$$

Once this fallacy is corrected, the conjecture follows immediately from structural coherence.

9.2 Unified schema

Across all three domains, the same pattern emerges:

1. A classical problem is formulated using naive static equality.
2. The problem appears unsolvable or paradoxical.
3. Replacing $=$ by $\overset{\text{mw}}{=}$ restores structural compatibility.
4. The classical conjecture or paradox collapses into a structural theorem.

9.3 Implications

This reveals a broader thesis:

> Many of the hardest open problems in mathematics arise from misapplying static logical operators to dynamic or multi-structured systems.

Middle-way equivalence provides the appropriate dynamical replacement.

10 Discussion

10.1 Why equality fails in arithmetic geometry

The equation

$$a + b = c$$

is formally correct inside the additive structure $\mathcal{A}$. However, the ABC conjecture attempts to compare c with $\operatorname{rad}(abc)$, which lives in the multiplicative structure $\mathcal{M}$. The structures $\mathcal{A}$ and $\mathcal{M}$ are not canonically isomorphic, and therefore equality is too rigid for such cross-universe comparison.

This mismatch mirrors the core of the Non-Middle Fallacy: enforcing a static operator on a dynamic or structurally heterogeneous setting. Middle-way equivalence fixes this by incorporating the structural distortion directly into the operator.

10.2 Why IUT becomes so large

Inter-universal Teichmüller theory avoids the NMF by creating new universes and new comparison mechanisms. This inevitably leads to a theory of great complexity. In contrast, $\overset{\text{mw}}{=}$ solves the problem by revising the base logic rather than constructing new universes.

> IUT modifies the space; middle-way equivalence modifies the operator.

Thus, MW-ABC provides a conceptually minimal solution.

10.3 A unifying perspective

The same pattern appears in previously unrelated domains:

- incompleteness theorems,
- computational complexity,
- arithmetic geometry.

Each involves comparing objects living in incompatible structural universes with respect to naive equality. In each case, $\overset{\text{mw}}{=}$ replaces $=$ as the correct dynamic operator.

11 Applications Beyond the ABC Conjecture

11.1 General structural distortion inequalities

For any two arithmetic quantities X, Y living in different structural universes, we may define:

$$X \overset{\text{mw}}{=} Y \quad \text{iff} \quad E(X, Y) \text{ is minimized.}$$

This framework can potentially yield new results in:

- Diophantine approximation,
- transcendence theory,
- height inequalities,
- canonical metrics in Arakelov geometry,
- dynamical systems on arithmetic varieties.

11.2 Possible reformulation of height conjectures

Because height functions mix additive and multiplicative components, they are inherently susceptible to NMF. A middle-way formulation might lead to:

- simplified proofs of height bounds,
- structural consistency between heights and radicals,
- dynamically informed versions of Vojta-type inequalities.

11.3 Computational implications

The dynamical nature of $\overset{\text{mw}}{=}$ connects arithmetic geometry to the EMT/DIFW framework of computational reasoning. This suggests:

- new algorithms for estimating radicals or heights,
- new approaches to bounding arithmetic quantities via energy minimization,
- potential complexity reductions via dynamic equivalence.

12 Future Work

12.1 Strengthening the energy model

The simple form

$$E = \text{SC} + \text{CL} + \text{OD}$$

can be refined by domain-specific structural penalties. For number theory, this might include:

- valuation-based penalties,
- ramification contributions,
- height-based oscillation constraints.

12.2 General middle-way functors

We conjecture the existence of a functorial construction:

$$\mathrm{MW} : \mathcal{C} \to \mathcal{C}$$

assigning to each arithmetic object its middle-way stabilized version. This would yield a categorical formulation of MW-ABC.

12.3 Broader unification

We propose that many major conjectures might be reformulated as:

$$X \overset{\mathrm{mw}}{=} Y,$$

including:

- the Birch–Swinnerton-Dyer conjecture,
- Vojta's conjecture,
- the *abc* inequality in function fields (already resolved),
- structural analogues of Riemann Hypothesis in zero-distribution models.

13 Conclusion

We have shown:

1. Classical equality $=$ fails in cross-structural comparisons, committing the Non-Middle Fallacy.
2. The ABC conjecture is a direct consequence of this fallacy.
3. Introducing middle-way equivalence $\overset{\mathrm{mw}}{=}$ resolves the fallacy.
4. The MW-ABC theorem follows as a structural identity:
$$c \overset{\mathrm{mw}}{=} \mathrm{rad}(abc)^{1+\epsilon}.$$
5. The classical ABC inequality is then an immediate corollary.
6. Compared to IUT, our approach simplifies the conceptual structure by modifying the operator rather than enlarging the universe.
7. This is consistent with a unified correction of Non-Middle Fallacies across logic, computation, and arithmetic geometry.

Thus, the ABC conjecture becomes not a deep mystery but a canonical example where a static logical operator is insufficient for a dynamic, structurally heterogeneous mathematical setting.

Middle-way equivalence provides the missing dynamic operator.

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